Radio Programming

Program a radio from a current, authorized channel plan, then verify the saved settings. Use a separate guide for your exact radio model, software, and connection method.

△ **CAUTION — PREVENT HARM**

Confirm compatibility first. Being able to enter a frequency does not authorize transmission. The radio, operator, and settings must be permitted for the radio service and channel. Ask the radio lead if any requirement is unclear.

GET THE CHANNEL PLAN FROM THE RADIO LEAD

Setting	What to confirm
Channel name and memory	Agreed label and destination slot; avoid overwriting a needed channel.
Receive frequency (RX)	Frequency your radio listens to.
Transmit frequency (TX)	Frequency your radio sends on. Simplex: same RX/TX. Repeater: may differ.
Repeater shift or offset	Confirm direction and amount if required; check the resulting TX frequency.
Tone or code	Confirm TX and RX separately: none, CTCSS, or DCS; include DCS polarity (normal/inverted) when specified.
Mode, bandwidth, power	Exact plan requirements and radio compatibility; use authorized power appropriate to the link.

CTCSS and DCS filter which received signals open the speaker. They do not encrypt a conversation or make a channel private.

LOAD AND SAVE THE SETTINGS

1. **Identify the exact model and software.** Follow its guide for keypad entry, a cable, or an app. Do not substitute another model's button sequence.
2. **Back up.** Read and save the existing configuration when supported; confirm you connected to the intended radio.
3. **Enter the channel plan.** Set the name, RX, TX, tones/codes, mode, bandwidth, and power. Check decimal places, units, and DCS polarity.
4. **Save to the radio.** Complete the model's write or synchronization step. An entry saved in phone or computer software may not yet be stored in the radio.
5. **Read back the stored channel.** Reopen or read from the radio and compare each value with the plan. Stop if any field is missing or does not match.

SOURCES

SAMPLE TEXT — NOT APPROVED ADVICE. Sources: Fire Watch programming brief; COM-008.

Verify a Programmed Radio

A successful write is only the first check. Verify that the radio retains the channel and can communicate as intended.

CHECK BEFORE RETURNING TO SERVICE

- ☐ Restart as directed; confirm the saved channel remains available.
- ☐ Select the intended memory/group; check volume and receive audio.
- ☐ Check the active transmit channel when using dual watch or scanning.
- ☐ Verify any talk-around or reverse setting against the plan; these can change the transmit path.
- ☐ When authorized and the channel is clear, make a short radio check with a known station. Follow COM-002 for call procedure.
- ☐ Confirm both stations can hear and understand each other; a receive-only check does not verify your transmit settings.
- ☐ Record the plan version, radio, date, and result. Report unresolved problems to the radio lead before duty.

RECORD THE VERIFIED CHANNEL PLAN

Complete from the authorized plan. Keep the record with the radio's setup instructions.

Field	Confirmed value
Radio model and identifier	_____
Channel plan version and date	_____
Channel name, memory, and group	_____
RX frequency and TX frequency	_____
Shift / offset, if used	_____
RX and TX tones/codes, DCS polarity	_____
Mode, bandwidth, and power	_____
Programmed by and date	_____
Two-way check with station and result	_____

USE THE MATCHING MODEL GUIDE

COM-008: BTECH UV-PRO Getting Started covers that model's controls, pairing, and channel entry. Keep menus, cables, apps, and screenshots for other radios, such as the Baofeng UV-5R Mini, in separate guides verified for their exact model and firmware.

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